

Entity/Relationship Definition and Modeling Specification

1. Purpose

This specification defines what the project means by entity, relationship, knowledge graph, and CMOC subgraph. It also defines the architecture, tools, algorithms, output schema, and code changes needed to make entity/relationship modeling scientifically defensible.

This document is different from `Entity_Relationship_Verification_Protocol.md`.

- The verification protocol explains how to score entity and relationship outputs.
- This specification explains what entities and relationships are supposed to be before they are scored.

The document directly answers the professor's questions:

1. What is a node?
2. What is an edge?
3. Does an edge have direction?
4. Does an edge have a name or relation type?
5. Are relation types predefined or generated by the model?
6. Where will entity and relationship outputs be stored so that humans can read and evaluate them?

2. Executive Summary

In this project, entities are not arbitrary keywords. Entities are Richmond-relevant "big concepts" that function as Context, Intervention, Mechanism Resource, Mechanism Response, or Outcome concepts in a realist synthesis.

Relationships are not visual lines between circles. Relationships are directed, typed, evidence-supported claims that connect two normalized entities in a way that supports a CMOC or programme theory statement.

The official graph is therefore defined as a controlled, evidence-backed, Richmond-aligned graph:

```
Entity = normalized Richmond-relevant concept
Relationship = directed typed evidence-backed link between two entities
CMOC = small graph motif connecting Context, Intervention, Mechanism, and Outcome
Programme Theory = cross-study pattern made from recurring CMOC motifs
```

3. Research Foundation

Richmond et al. (2020) use realist synthesis. In that framework, evidence is interpreted as Context-Mechanism-Outcome configurations. Contexts affect how an intervention is received. Mechanisms include resources offered by the intervention and participant responses triggered by those resources.

Outcomes are the effects or consequences of those responses.

This maps naturally to a knowledge graph because the core review product is relational:

Context conditions how an Intervention/Resource triggers a Mechanism Response, which leads to an Outcome.

GraphRAG is relevant because it constructs a graph index from documents, partitions the graph into communities, and uses community summaries for global sensemaking across a corpus (Edge et al., 2025). This supports the project's need to reason over the entire Richmond corpus rather than retrieve isolated passages.

The project abstract frames the Literature Knowledge Graph as an external evidence base built from entities and relations, where entities correspond to constructs such as interventions, populations, outcomes, contexts, study designs, and assessments, and relations capture how those constructs are linked.

4. Entity Definition

An entity is a stable, review-relevant concept that can participate in a realist explanation.

An entity must satisfy all of the following conditions:

1. It is relevant to Richmond's review question.
2. It belongs to one of the official CMOC categories.
3. It has textual or synthesis evidence.
4. It can be mapped to an E-code, or it is explicitly marked as a human-approved new concept.
5. It is not merely a surface phrase, citation, author name, journal name, or isolated methodological detail.

5. Official Entity Categories

The official entity categories are:

Category	Meaning	Gold code range
Context	Learner, group, condition, or setting factor that changes how an intervention works.	E01-E06
Intervention	Pedagogical activity, instructional feature, learning environment, or feedback process.	E07-E18
Mechanism Resource	Resource or opportunity offered by an intervention.	E19
Mechanism Response	Cognitive, affective, reasoning, coping, or behavioral response triggered in the learner.	E20, E23, E24, E26, E27, E31-E35, E39, E42, E45
Outcome	Learning, reasoning, diagnostic,	E21, E22, E25, E28-E30, E36-

Category	Meaning	Gold code range
	ffective, or future-performance consequence.	E38, E40, E41, E43, E44, E46, E47

Local implementation evidence:

- `data/gold_standard.json` defines E01-E47.
- `core/ontology.py` defines the corresponding Pydantic entity categories and label enums.

6. Entity Levels

The project maintains three entity levels:

Level	Description	Example use
Raw entity	Phrase or concept extracted by GraphRAG or an LLM from source text.	Candidate node list.
Normalized entity	Raw entity mapped to a controlled E-code or rejection category.	Evaluation and CMOC construction.
Gold entity	Richmond-derived reference concept in E01-E47.	Benchmark comparison.

This distinction prevents a common error: treating every generated graph node as automatically correct.

7. Entity Output Schema

Every official entity row should contain:

```
entity_instance_id
raw_label
normalized_e_code
gold_label
category
source_document_id
evidence_text_or_location
evidence_level
confidence
human_status
notes
```

Recommended `human_status` values:

```
accepted
revised
rejected
unresolved
```

Recommended `evidence_level` values:

```
full_text
abstract_only
richmond_synthesis
derived_from_human_adjudication
```

8. Relationship Definition

A relationship is a directed, typed, evidence-supported link between two normalized entities.

The official relationship unit is:

```
(source_e_code, predicate, target_e_code)
```

Example:

```
(E13, TRIGGERS, E42)
```

Meaning:

Accurate and timely feedback triggers learners to develop understanding of their successes and failures and generate plans for improvement.

9. Official Relationship Vocabulary

For strict Richmond evaluation, the relation vocabulary is frozen to the predicates used in `data/gold_standard.json`:

Predicate	Meaning
PROVIDES	An intervention supplies a resource, opportunity, case, explanation, or feedback.
ENABLES	A context or condition makes a mechanism possible or more likely.
TRIGGERS	A resource or intervention activates a learner response.
LEADS_TO	A mechanism response produces or contributes to an outcome.
CONSTRAINS	A context, missing resource, or negative response limits, blocks, or worsens an outcome.

Implementation implications:

- **CONSTRAINS** must be added to `core/ontology.py`, or all **CONSTRAINS** edges must be explicitly mapped before evaluation.
- **INHIBITS** is treated as a relaxed alias of **CONSTRAINS**, not as a separate strict gold predicate.
- **MODERATES** may be retained as an analytic note for broader theory discussion, but it does not replace the strict Richmond R01-R40 predicate set unless the gold file is revised.

10. Relationship Direction Rules

Direction matters. A reversed edge is not equivalent.

Pattern	Direction
Intervention provides resource	Intervention -> Mechanism Resource
Intervention triggers response	Intervention -> Mechanism Response
Resource triggers response	Mechanism Resource -> Mechanism Response
Context enables response	Context -> Mechanism Response
Context constrains outcome	Context -> Outcome
Response leads to outcome	Mechanism Response -> Outcome

The resulting edge policy is:

Yes. In the official graph, every edge must have direction, predicate name, evidence, and source/target E-code endpoints.

11. CMOC as a Graph Motif

The smallest useful graph pattern for this project is not a single edge. It is a CMOC motif:

```
Context
-> enables/constrains
Intervention or Mechanism Resource
-> triggers/provides
Mechanism Response
-> leads_to
Outcome
```

The current code already approximates this through `SingleCMOC` in `core/ontology.py`:

```
context
intervention
mechanism_resource
mechanism_response
outcome
causal_chain
programme_theory_statement
is_negative
confidence
```

The next implementation step is to ensure that each CMOC's `causal_chain` is exported as strict E-code triples and compared to R01-R40.

12. Proposed Architecture

The entity/relationship modeling architecture follows this sequence:

Step	Tool/component	Output
1. Document registry	<code>studies_metadata.json</code> l; extracted PDF/abstract text	28-study input table

Step	Tool/component	Output
2. GraphRAG indexing	Microsoft GraphRAG-style indexing	candidate entities, relationships, communities, community reports
3. Entity normalization	ontology mapper using E01-E47	entity mapping table
4. Relationship materialization	triple exporter	(source_e_code, predicate, target_e_code) table
5. CMOC construction	realist synthesis plugin / Pydantic schema	per-paper CMOC motifs
6. Evaluation	entity/relationship/CMOC evaluator	metrics and error categories
7. HITL refinement	human review tables	accepted/revised/rejected corrections
8. Synthesis	cross-study synthesizer	PTS1-PTS5 and final programme theory

13. Algorithms and Methods

The project should use a hybrid approach:

13.1 GraphRAG for candidate graph structure

GraphRAG constructs an entity graph from source documents, detects communities, and generates community summaries for global sensemaking across a corpus. Edge et al. (2025) specifically describe hierarchical community detection using Leiden and community-level summaries.

Use in this project:

- identify candidate concepts and relations across the 28-study corpus;
- generate community summaries that reveal recurring conceptual clusters;
- support global questions such as "what are the dominant contexts and mechanisms across the review?"

13.2 Richmond ontology for controlled normalization

GraphRAG candidate nodes are not treated as final. They must be normalized to the Richmond-derived E01-E47 ontology.

Use in this project:

- convert raw graph concepts into Richmond-aligned entities;
- reject unsupported or irrelevant nodes;
- preserve human-approved new concepts separately.

13.3 LLM extraction with constrained schema

The CMOC extractor should use a constrained schema rather than free-form summaries. The current Pydantic schema in `core/ontology.py` is the right direction because it limits outputs to Context, Intervention, Mechanism Resource, Mechanism Response, Outcome, and typed relationships.

Use in this project:

- extract structured CMOC motifs per study;
- require evidence quotes;
- assign PTS1-PTS5;
- mark negative pathways.

13.4 Human-in-the-loop adjudication

Because Richmond does not provide a complete per-paper machine-readable CMOC gold set, humans must adjudicate the uncertain or high-impact outputs.

Use in this project:

- approve or revise entity mappings;
- approve or revise relationship triples;
- label error types;
- feed corrections back into prompts, schemas, or mapping tables;
- rerun and re-score affected stages.

14. Required Output Files

The entity/relationship subsystem produces these human-readable files:

File	Purpose
outputs/entity_mapping_table.csv	Raw nodes mapped to E01-E47 with evidence and human status.
outputs/relationship_triples.csv	Directed E-code triples mapped to R01-R40.
outputs/kg_verification_report.md	Entity and relationship metrics.
outputs/per_paper_cmoc_adjudication_template.csv	Human adjudication template for CMOCs, entities, and relationships.
outputs/error_analysis.csv	Error categories used for refinement.

15. Current Implementation Gaps

Gap	Evidence	Fix
CONSTRAINS missing from RelationType	data/gold_standard.json uses it; core/ontology.py does not.	Add CONSTRAINS or map it explicitly.
Relationship outputs use local IDs	outputs/relationships_long.json uses IDs like i1, mr1, o1.	Export E-code triples.
Relationship F1 not computed	scripts/evaluate_phase1.py currently computes E-code/PTS/chain metrics but not strict R01-R40 edge scoring.	Add relationship precision, recall, F1, direction, and predicate accuracy.
HITL not yet explicit	core/orchestrator.py includes checkpoints but current run logs auto-approval behavior.	Add reviewable tables and human decision logs.
Corpus indexing mismatch	graphrag/input/ has 28 text files, but GraphRAG stats report 20 indexed documents.	Audit and rebuild or explain index coverage.

16. Summary Position

The graph nodes are Richmond-aligned realist synthesis concepts, not arbitrary keywords. They are normalized to Context, Intervention, Mechanism Resource, Mechanism Response, and Outcome entities using E01-E47. The graph edges are directed, typed, evidence-supported relationships using a frozen predicate set: PROVIDES, ENABLES, TRIGGERS, LEADS_TO, and CONSTRAINS. Each edge must be exported as an E-code triple and compared with R01-R40. GraphRAG helps discover candidate graph structure and communities, but the final research graph is controlled by the Richmond ontology and verified through human-in-the-loop adjudication.

17. References

Richmond, A., Cooper, N., Gay, S., Atiomo, W., & Patel, R. (2020). The student is key: A realist review of educational interventions to develop analytical and non-analytical clinical reasoning ability. *Medical Education*, 54(8), 709-719. <https://doi.org/10.1111/medu.14137>

Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R. O., & Larson, J. (2025). From local to global: A GraphRAG approach to query-focused summarization. arXiv:2404.16130v2.

Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From Louvain to Leiden: guaranteeing well-connected communities. *Scientific Reports*, 9(1).

Professor project abstract. (2026). Proposed AI Agent Pipeline for Systematic Review in Education.

Local file: documents/abstract-from-professor/abstract.md.

Project gold standard. (2026). Richmond-derived E01-E47, R01-R40, and PTS1-PTS5 benchmark.

Local file: data/gold_standard.json.